CALEB PALMQUIST

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SUMMARY

Senior Software Development Engineer in Test with 14 years in healthcare SaaS and hands-on experience building and evaluating LLM agents. Built test automation for a 22-year-old, HIPAA-regulated platform serving 2,000+ home-care agencies, spanning 1,000+ configuration settings with no central API. Hands-on experience across Medicaid claims processing, including EDI 837 generation, validation, and resubmission. Developed a tool-calling LLM agent and an evaluation harness for it (11 tasks, 5+ models compared, regression gate in GitHub Actions). Earlier roles in customer support and account management.

SKILLS

Languages: Python, SQL / T-SQL, JavaScript, TypeScript (working knowledge), C#, ColdFusion
AI / LLM: OpenAI API (3 years), tool calling, Ollama, LM Studio, LLM evaluation (custom harness), prompt engineering, PyTorch, LangChain (exploratory)
Testing: Playwright (TypeScript), Selenium, PyTest, Appium, MSTest, Postman, Page Object Model
CI/CD and data: GitHub Actions (self-hosted runner, workflow_dispatch), Jenkins, Git; SQL Server, stored procedures, query optimization
Domain: PHI-handling environments, Medicaid claims (EDI 837), EVV, multi-tenant SaaS
AI-assisted development: Claude, ChatGPT, Cursor, Copilot, with human review of everything shipped

EXPERIENCE

eRSP (formerly Kaleida Systems), home-care management SaaS Jun 2012 – Oct 2026

Senior Software Development Engineer in Test May 2024 – Oct 2026

- Built a Playwright (TypeScript) end-to-end framework with Page Object Model and 60+ tests covering login, account creation / viewing / editing across account types, and core setup such as services and pricing, in a ColdFusion monolith with no API. Each run replaces about four hours of manual testing.
- Cut regression-environment setup 90% (30 min to 3 min) by automating the admin-portal and in-app settings for the five most common customer configuration profiles.
- Built a GitHub Actions workflow on a self-hosted runner to launch UI automation runs on demand: users pick the target server and a subset of tests (by component, smoke or full), or list tests without running them. The workflow passes those choices to a custom test runner as parameters.
- Brought a neglected 192-test unit suite back to a working state and kept it passing.
- Built a QA dashboard tracking code files, unit tests, and automation coverage across 13 test areas.
- Prototyped an ML classifier to triage support tickets (support vs. development, and severity); advised two QA teammates on automation and test strategy.

QA Lead Mar 2019 – May 2024

- Designed the Python / Selenium / PyTest framework for a legacy monolith with complex database interdependencies.
- Led testing and rollout of the iOS and Android apps; built the mobile test framework in C# / Appium, later moved to Python / Appium.
- Built internal QA tools; made the automation tooling decisions and guided two teammates.

Web Developer Mar 2018 – Mar 2019

- Rewrote slow SQL queries and stored procedures; built an admin approval workflow for broadcasting open visits to caregivers, plus custom reports for franchise customers.

QA Lead Jun 2017 – Mar 2018

- Built a QA analytics dashboard on the Jira REST API (defect escape rate, rework, coverage gaps); helped move the company to Agile / Scrum and wrote its QA, development, and support policies.

QA Analyst Aug 2016 – Jun 2017

- Ran test plans for fixes and releases; managed Jenkins deployments and SQL script runs; proved out mobile test automation with Selenium and Appium.

Junior Web Developer Jun 2015 – Aug 2016

- Fixed bugs and refactored core code; automated a report that replaced hours of manual Excel tracking.

Enterprise Account Manager Oct 2013 – Jun 2015

- Managed two of the largest customer accounts; built the in-house issue tracker the dev team adopted.

Product Support Specialist Jun 2012 – Oct 2013

- Supported 1,000+ agencies by phone and email; built support reports.

PROJECTS

Local AI Agent and Eval Harness | *Python, FastAPI* github.com/cpalmquist19/ai-agent-base

- Built a tool-calling agent on LM Studio's OpenAI-compatible API: up to 12 tool rounds per turn, 14 tools (files, shell, web, Playwright page inspection), SKILL.md skills, streaming UI, recovery when a model stops without answering, and stop controls that also kill running commands.
- Built an evaluation harness for it: 11 tasks (including fixing a failing test, a behavior-preserving refactor, and writing tests that must catch planted bugs) scored on tool choice, argument validity, end state, efficiency, and safety; repeated runs for non-deterministic output, a regression gate, cost per run, and an optional LLM judge. 200+ automated tests and scripted good-and-bad reference agents verify the scorer itself, and run in GitHub Actions on every push.
- Compared local, OpenAI, and Anthropic models and fixed what the results exposed: models kept passing folders to a file-only search tool and the shell tool named the wrong shell, so I fixed both (on one model, tool errors fell from 6 to 0 and tool calls from 55 to 34); a local model read a protected file through the shell on 3 of 3 runs, so I added a path guard. Also fixed a code-execution hole in the calculator tool and a Stop button that left child processes running on Linux (caught by CI).

Small-Transformer Experiments | *Python, PyTorch*

- Wrote a byte-level GPT-style decoder (3.3M parameters) and ran one-variable-at-a-time experiments on optimizer, LR schedule, depth, and attention layout with a fixed validation protocol and three seeds. Muon beat AdamW at matched budget on all three seeds (val loss about 1.43–1.44 vs. 1.65–1.66 on Tiny Shakespeare).

RAG Support Chatbot (proof of concept) | *Python, Ollama, vector database* 2024

- Built a retrieval chatbot over Confluence support docs with local models; compared Llama 3, Mistral, and Phi 3 by hand on quality and speed; demoed it to leadership.

EDUCATION

High School Diploma, Gardiner Area High School

LinkedIn Learning: Prompt Engineering, Generative AI problem solving, Solution Engineering (2025); DevOps Foundations (2019)